

Thermal Reserve

Multi-day, snapback-aware thermostat setbacks that keep Southcentral Alaska's gas demand under the line on the coldest days.

Try it on your phone: thermal-reserve.vercel.app/home



THE PROBLEM

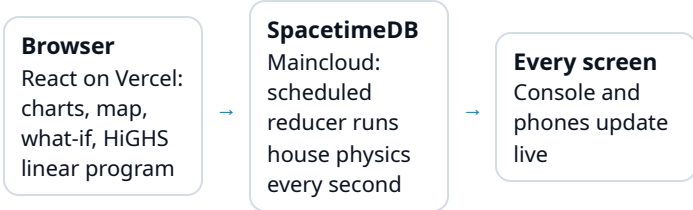
- Enstar's president told lawmakers the utility could enter this winter about **3 Bcf short**, about **18 days** of winter supply. sourced
- Cook Inlet storage held **~6.54 Bcf** on Aug 7, 2026, of 13 Bcf working capacity. sourced
- In a shortage, Enstar would ask **large commercial and industrial users to cut first**, before homes. LNG imports may not be ready until **late 2029**. sourced
- Short thermostat events elsewhere mostly shift gas use later: ConEd measured **52%** of savings lost to snapback. sourced

WHAT WE BUILT

A **simulation**, not real thermostat control. An operator console plans and dispatches setbacks across a simulated fleet of Anchorage homes (24 home types, 1,000 map dots), solving a linear program for the least discomfort that keeps each gas day under capacity.

Households join from a phone, see indoor temperature and gas saved, and can override in one tap. A **62°F comfort floor** (never below 60°F) is enforced on the server; households needing steady heat are exempt.

HOW IT WORKS



One physics file runs in the browser and on the server; a test checks they match.

HONEST IMPACT

Sustained setback at -20°F, automated homes only. derived UA 398.3 BTU/(h·°F) from Enstar's 149 Mcf average home, 30 years of Anchorage degree-days and 988 BTU/cf gas (EIA); efficiency 0.85 assumed

Enrolled homes	5°F setback	Share of 20 MMcf/day needle peak	Value at \$17.50/Mcf
10,000	0.57 MMcf/day	2.8%	~\$10,000/day
25,000	1.42 MMcf/day	7.1%	~\$24,900/day
50,000	2.85 MMcf/day	14.2%	~\$49,800/day

What it doesn't do: fix the seasonal shortfall. Over 20 cold days, 25,000 homes save ~28 MMcf, under 1% of 3 Bcf. It's a deliverability tool for the worst days.

VALIDATION

Same physics, run under published pilot conditions. derived

Test	Published	Ours	Band	Result
ConEd-like snapback	retention 0.48	0.465	0.38–0.58	Pass
SoCalGas-like daily	2.2%	1.14%	1.5–3.0%	Gap

The gap is shown, not tuned away: the two pilots imply different snapback, and our model errs low on daily savings.